

Determinants



2x2 determinants

- 1 Find the determinant of the matrix with rows $\langle 4, 3 \rangle, \langle 2, 5 \rangle$.
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3x3 determinants via cofactor expansion

- 2 Find the determinant of the matrix with rows $\langle 1, 2, 3 \rangle, \langle 0, 1, 4 \rangle, \langle 5, 6, 0 \rangle$ by cofactor expansion along the first row.
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The determinant product rule

- 3 For A with rows $\langle 1, 2 \rangle, \langle 3, 4 \rangle$ and B with rows $\langle 2, 1 \rangle, \langle 0, 3 \rangle$, verify $\det(AB) = \det(A)\det(B)$.
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The invertibility criterion

- 4 State the invertibility criterion in terms of the determinant.
 - 5 Is the matrix with rows $\langle 2, 4 \rangle, \langle 1, 2 \rangle$ invertible? Justify using the determinant.
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Applying the invertibility criterion

- 6 For what value(s) of k is the matrix with rows $\langle k, 2 \rangle, \langle 3, k \rangle$ NOT invertible?